Mini Assessment

Name : Kola.Srikanth
Reg No : 99230040341
Dept : CSE
Date : 11-09-2026

Q1. If a database connection times out, what should your program do?

A) A connection timeout means the database server did not respond within the allowed time, often due to network issues, server overload, or an unreachable host.

→ Catch the specific timeout exception (e.g. OperationalError) instead of letting the program crash.

→ Retry the connection a limited number of times using a short backoff delay between attempts.

→ Set a sensible connection timeout value so the program does not hang indefinitely.

→ If retries are exhausted, log the failure and alert the user with a clear, non-technical error message.

Q2. If cursor.fetchone() returns None, what should your program do?

A) cursor.fetchone() returns None when a query executes successfully but matches no rows, for example when searching for a record that does not exist in the table.

The program should always check the result before using it, rather than assuming a row was returned. Trying to index or unpack None directly raises a TypeError and crashes the program. Instead, wrap the access in a conditional check, return a clear "not found" response or default value, and log the query context so the missing-record case can be investigated if needed instead of failing silently or with an unhandled exception.

3. If a column value is missing from a query result row, why might .get() be safer than []?

A) When a database row is returned as a dictionary (e.g. via a DictCursor or RealDictRow), accessing row['column'] raises a KeyError if that column was not selected or is absent from the result. row.get('column') returns None instead, making it safer for optional or dynamically selected columns.

→ .get() prevents unnecessary KeyError exceptions.

→ A default value can be supplied: row.get('email', 'Unknown').

→ It is useful when query results may have optional or dynamically included columns.

→ The program can check for None and decide what to do next.

→ This improves reliability when working with changing schemas or partial SELECT statements.

Example: If a customer's middle name is missing, row.get('middle_name', 'Not provided') safely supplies a default value.